

SAMUEL ZHENG

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EDUCATION

Bachelor of Computer Science, *University of Waterloo*

Expected April 2027

EXPERIENCE

Software Engineer Intern | *Java, Spring Boot, TypeScript, Flutter, MCP*

NimbleRx

January – April 2026
Redwood City, California

- Deployed AI-automated workflows using **Claude Code** and **Model Context Protocol (MCP)**, automating technical design reviews and E2E testing to accelerate sprint velocity by **20x**.
- Architected a backend refactor of **Java** services using **Domain-Driven Design (DDD)**, reducing code complexity metrics and increasing system scalability for autonomous agentic workflows.
- Spearheaded forward-deployed engineering for refill monetization, designing and implementing projects that reduced churn risk by **95%** for high-value Pharmacy/Pharma clients.
- Developed and maintained mission-critical features across **TypeScript/React** web portals and **Flutter** mobile apps, improving the prescription fulfillment UX for thousands of pharmacy partners and millions of patients.

Software Engineer Intern | *Java, Spring Boot, GraphQL, PostgreSQL*

NimbleRx

May – Aug 2025
Redwood City, CA

- Architected a unified **GraphQL API layer** to consolidate **SQL** and **RESTful** endpoints, achieving a **80% reduction in query latency** for distributed pharmacy systems.
- Engineered high-throughput **Java/Spring microservices** backed by **PostgreSQL**, processing **1M+ daily transactions** while reducing database IOPS by **30%** through query optimization.
- Developed a **FHIR/HIPAA-compliant** medical record lookup system for 6,000+ pharmacies, unlocking a new B2B revenue stream via secure, real-time data exchange.

Security Software Developer Intern | *Python, TypeScript, React, AWS*

ISARA Corporation

Sep – Dec 2024
Waterloo, ON

- Automated regression testing workflows using **Python** and **Bash**, integrating system metrics into a unified dashboard for real-time performance bottleneck analysis.
- Refined a **Post-Quantum Cryptographic** inventory tool using **TypeScript, React, and GraphQL**, resolving **90% of legacy technical debt** and improving UI responsiveness.
- Deployed scalable, containerized infrastructure on **AWS (EC2, S3, IAM)** using **Docker**, strengthening system reliability and granular access control.

Product Solutions Engineer Intern | *Python, Azure, TensorFlow*

Advanced Micro Devices Inc. (AMD)

Jan – Apr 2024
Markham, Ontario

- Led development of automated **GPU benchmarking pipelines**, applying **machine learning models** to analyze performance metrics on video games and graphics-intensive software, cutting manual validation time by **100%**, and increasing test volume by **10x**.
- Built full-stack **Python/Express** dashboards for **100+ engineers**, accelerating report generation and improving visibility into GPU performance data.

PROJECTS

printbnb | *Next.js, React, Firebase, Figma*

printbnb.ca

- Co-founded a P2P marketplace for 3D printing resources, scaling to **100+ active users** and handling end-to-end product delivery from concept to deployment.
- Implemented real-time messaging and order state machines using **Firebase Cloud Functions** and **Firestore**.

SKILLS

Languages: Java, C/C++, Python, JavaScript, TypeScript, Kotlin, Flutter, Bash, HTML/CSS
Frameworks/DBs: Spring Boot, PostgreSQL, MySQL, GraphQL, REST, Firebase, Express, Node.js
Infrastructure: AWS (EC2, S3, IAM, CloudWatch), Docker, Kubernetes, GCP, Linux, CI/CD
AI/Tools: Claude Code, MCP, Cursor, LLM Fine-tuning, Postman, Git